

TECHNICAL PROGRAMME

NEW2AN/ruSMART 2021

Online participation by Youtube + Zoom



Monday, August 30, 2021

9:20 – 9:30 Opening. Welcome words from Prof. Yevgeni Koucheryavy

9:30 – 10:20 Keynote talk: Dr. Evgeny Khorov

10:20 – 10:30 Q&A for the Keynote Talk: ZOOM session

10:30 – 11:50 Section I: ruSMART I

Fast data processing by IoT devices

Alexander Y Anufrienko (Higher School of Economics, Russia)

On Smart Greenhouse Issues

Manfred Schneps-Schneppe (Ventspils University College Ventspils International Radio Astronomy Centre, Latvia); Gunars Lacis (Institute of Horticulture, Latvia)

Smart Contract Enabled Decentralized Reputation System for E-commerce Reviews

Carl Kugblenu, Petri Vuorimaa and Barbara Keller (Aalto University, Finland)

The estimation of Quality of Service in Tactile Internet, Augmented Reality and Internet of Things

Abbas Alzaghir (The Bonch-Bruевич Saint-Petersburg State University of Telecommunications, Russia); Andrey Koucheryavy (SPbSUT, Russia); Alexander Paramonov (The Bonch-Bruевич Saint-Petersburg State University of Communication, Russia)

The Challenges with Internet of Things Security for Business

Ievgeniia Kuzminykh (King's College London, United Kingdom (Great Britain) & Kharkiv National University of Radio Electronics, Ukraine); Bogdan Ghita (University of Plymouth & Centre for Security, Communications, and Network Research, United Kingdom (Great Britain)); Jose Such (King's College London, United Kingdom (Great Britain))

Information and communications technology in the development of territories based on designing "smart cities"

Viktoria Bondarenko (Rostov state University of Economics, Russia); Natalia Guzenko and Tatiana Romanishina (Rostov State University of Economics, Russia); Valery Leventsov (Peter the Great St. Petersburg Polytechnic University, Russia); Vladimir Gluhov (Saint-Petersburg State Polytechnical University, Russia)

11:50 – 12:00 Q&A for Session I: ZOOM session

12:00 – 13:20 Section II: Business Informatics I

Developing smart cities: the risks of using information and communications technology

Viktoria Bondarenko (Rostov state University of Economics, Russia); Tatiana Romanishina and Natalia Guzenko (Rostov State University of Economics, Russia); Natalya Mukhanova and Sergey Salkutsan (Peter the Great St. Petersburg Polytechnic University, Russia)

Influence of digital technology and telecommunications on the customer-oriented development of electronic commerce

Olga Chkalova (Lobachevsky State University of Nizhni Novgorod, Russia); Yury Trifonov and Pavel Shalabaev (Lobachevsky State University of Nizhny Novgorod, Russia); Ekaterina Abushova (Peter the Great Saint-Petersburg Polytechnic University, Russia); Elena Kasianenko (Peter the Great St. Petersburg Polytechnic University, Russia)

Utilization of Organizational-Economic Mechanism for Selection and Management of Spectrum Sharing Scenarios to Increase Economic Efficiency of 5G operators

Valery Tikhvinskiy (Radio Research & Development Institute (NIIR) & Bauman Moscow State Technical University (BMSTU), Russia); Roman Umanskiy (Moscow Technical University of Communications and Informatics (MTUCI), Russia); Vladimir Makarov (The Bonch-Bruевич Saint Petersburg State University of Telecommunications, Russia); Arseny Plossky (Radio Research & Development Institute, Russia)

Sustainable Development of Small and Medium Business in View of the Rapid Growth of Telecommunications and Digital Economy in the Russian Federation

Marina Efremova, Maxim Tsvetkov and Nikolay Shimin (National Research Lobachevsky State University of Nizhny Novgorod, Russia); Oksana Anatolyevna Evseeva and Alexey Efimov (Peter the Great St. Petersburg Polytechnic University, Russia)

Info-communications-based interaction of companies and consumers on the grocery retail market

Olga Chkalova, Inna Bolshakova and Natalia Kopasovskaya (Lobachevsky State University of Nizhni Novgorod, Russia); Tatyana Nekrasova and Sergey Salkutsan (Peter the Great St. Petersburg Polytechnic University, Russia)

Transforming the strategic benchmarks of Russian telecommunications companies in the sustainable development paradigm

Irina Krasyuk and Oksana Anatolyevna Evseeva (Peter the Great St. Petersburg Polytechnic University, Russia); Maria Kolgan and Yulia Medvedeva (Don State Technical University, Russia)

13:20 – 13:30 Q&A for Session II: ZOOM session

13:30 – 14:50 Section III: Business Informatics II

Study of relationship between the corporate governance factors and ESG ratings of ICT companies from the developed markets

Sergei Grishunin (Higher School of Economics, Russia); Svetlana Suloeva (Peter the Great Saint-Petersburg Polytechnic University, Russia); Tatyana Nekrasova (Peter the Great St. Petersburg Polytechnic University, Russia); Alexandra Egorova (Higher School of Economics, Russia)

Improving Project Management for the Development of New Internet Applications

Tatyana Nekrasova (Peter the Great St. Petersburg Polytechnic University (SPbPU), Russia); Natalia Alekseeva (Peter the Great Saint-Petersburg Polytechnic University, Russia)

Specifics of forming an innovation sector when developing Industry 4.0 technology

Valery Leventsov (Peter the Great St. Petersburg Polytechnic University, Russia); Vladimir Gluhov (Saint-Petersburg State Polytechnical University, Russia); Anna Kamyshova and Denis Skripnichenko (St. Petersburg State University of Economics, Russia)

Increasing the competitiveness of info-telecommunications enterprises through building a mobile eco-system

Irina Krasnyuk, Valery Leventsov and Olga Kartavenko (Peter the Great St. Petersburg Polytechnic University, Russia); Maria Kolgan and Yulia Medvedeva (Don State Technical University, Russia)

Structural shifts on derivatives markets at the time of increasing digitalization and post-pandemic transformation of the market

Vladimir Gluhov (Saint-Petersburg State Polytechnical University, Russia); Olga Kartavenko (Peter the Great St. Petersburg Polytechnic University, Russia); Anna Kamyshova, Ekaterina Popova and Nikita Kapustin (St. Petersburg State University of Economics, Russia)

14:50 – 15:00 Q&A for Session III: ZOOM session

15:00 – 16:20 Section IV: Optics issues

Coarse estimation of the distance to the harmonic sound source by DAS for the determination of optical cable location

Vladimir Burdin (Povolzhskiy State University of Telecommunications and Informatics (PSUTI), Russia); Olga Gubareva and Vladimir Gureev (Povolzhskiy State University of Telecommunications and Informatics, Russia)

Fiber optic system for monitoring coolant parameters in nuclear power plants

Roman Davydov (Peter the Great St. Petersburg Polytechnic University, Russia); Semen Logunov and Denis Nikolaev (Peter the Great Saint Petersburg Polytechnic University, Russia); Vadim Davydov (Saint-Petersburg state Polytechnical University, Russia); Valentin Dudkin (The Bonch-Bruevich Saint - Petersburg State University of Telecommunications, Russia)

Fiber-optic sensor for monitoring radiation level

Diana S Dmitrieva (The Bonch-Bruevich Saint - Petersburg State University of Telecommunications, Russia); Valeria Pilipova (The Bonch-Bruevich Saint Petersburg State University of Telecommunications, Russia); Valentin Dudkin (The Bonch-Bruevich Saint - Petersburg State University of Telecommunications, Russia); Roman Davydov (Peter the Great St. Petersburg Polytechnic University, Russia); Vadim Davydov (Saint-Petersburg state Polytechnical University, Russia)

Experimental Study of Temperature Impact on Fiber Optic Current Sensor Elements

Valentina Temkina and Andrei Medvedev (Peter the Great St. Petersburg Polytechnic University, Russia); Alexey Mayzel (Software-Defined Systems LLC, Russia); Eduard Sivolenko (Radiophysics and Electronics Yerevan State University, Armenia); Ekaterina Poletaeva and Iuliia Dudnik (Peter the Great St. Petersburg Polytechnic University, Russia)

Fiber-Optic Recirculating Memory Loop for Wideband Microwave Signal

Sergei Ivanov, Alexander P. Lavrov and Dmitrii Kondakov (Peter the Great St. Petersburg Polytechnic University, Russia); Yuriy Matveev (Peter the Great Saint-Petersburg Polytechnic University, Russia)

16:20 – 16:30 Q&A for Session IV: ZOOM session

Tuesday, August 31, 2021

10:30 – 11:50

Section V: Drones and UAVs

Simulation and Experimental Study of Multi-Source Application Layer ARQ for FANET

Irina Kaisina (Kalashnikov Izhevsk State Technical University, Russia); Albert Abilov and Danil Vasiliev (Izhevsk State Technical University, Russia); Mohammed Amin Lamri and Anatoli Nistyuk (Kalashnikov Izhevsk State Technical University, Russia)

Deep learning approach for predicting energy consumption of drones based on MEC

Ali R. Abdellah (SPbSUT, Russia & Al-azhar University, Egypt); Abbas Alzaghir (The Bonch-Bruevich Saint-Petersburg State University of Telecommunications, Russia); Andrey Koucheryavy (SPbSUT, Russia)

Predicting energy consumption for UAV-enabled MEC using Machine Learning Algorithm

Abbas Alzaghir (The Bonch-Bruevich Saint-Petersburg State University of Telecommunications, Russia); Ali R. Abdellah (SPbSUT, Russia & Al-azhar University, Egypt); Andrey Koucheryavy (SPbSUT, Russia)

Section VI: Other topics

Investigation methods of dehydrated protein films for biomolecular electronics

Maksim Baranov (Peter the Great St. Petersburg Polytechnic University, Russia); Elena Velichko (Peter the Great St. Petersburg Polytechnic University, Russia)

Applying Deep Learning Techniques to Extract Diagnostic Information from ECG Images

Georgy Kostin (Peter the Great St. Petersburg Polytechnic University, Russia); Vitalii Pavlov (Peter the Great St Petersburg Polytechnic University, Russia)

Application of wavelet transform for ECG processing

Veronika Nikolaevna Malysheva, Diana Maratovna Zaynullina and Alena Olegovna Stosh (Peter the Great St. Petersburg Polytechnic University, Russia); Gregory Alexandrovich Cherepennikov (Saint-Petersburg State University of Aerospace Instrumentation, Russia)

11:50 – 12:00 Q&A for Session V and VI: ZOOM session

12:00 – 13:20 Section VII: Signals and PHY I

Analysis of nonlinear distortions of FTN signals transmitted through TWT amplifier

Ekaterina Smirnova and Sergey B. Makarov (Peter the Great St. Petersburg Polytechnic University, Russia)

Selecting a receiver for wideband spectrum sensing in cognitive radio systems based on an assessment of the signal environment complexity

Alexey Podstrigaev (Saint Petersburg Electrotechnical University LETI, Russia); Andrey Smolyakov (Saint Petersburg Electrotechnical University LETI & Scientific-Research Institute Vector OJSC, Russia); Vladimir Likhachev (Zhukovsky-Gagarin Air Force Academy, Russia); Sergei Efimov (Saint Petersburg State University of Aerospace Instrumentation, Russia); Vadim Davydov (Saint-Petersburg state Polytechnical University, Russia)

Instantaneous Interference Evaluation Model for Smart Antennas in 5G Ultra-Dense Networks

Grigoriy Fokin (The Bonch-Bruевич St. Petersburg State University of Telecommunications, Russia); Vadim Davydov (Saint-Petersburg state Polytechnical University, Russia); Vitaly Lazarev (The Bonch-Bruевич Saint - Petersburg State University of Telecommunications, Russia); Angelina Moroz (Peter the Great Saint Petersburg Polytechnic University, Russia)

The effect of error burst when using a decision feedback algorithm for receiving non-orthogonal multi-frequency signals

Sergey B. Makarov, Dac Cu Nguyen, Sergey V. Zavjalov and Anna S. Ovsyannikova (Peter the Great St. Petersburg Polytechnic University, Russia); Canh Minh Nguyen (University of Transport and Communications, Vietnam)

Software implementation of the algorithm for optimal joint estimation and detection of an arbitrary waveform

Nikita Ilchenko (Peter The Great Saint-Petersburg Polytechnic University, Russia); Evgenii Popov (Peter the Great St. Petersburg Polytechnic University, Russia)

Application of Neural Network as Demodulator of SEFDM Signals

Anastasiia Semenova (St. Petersburg Polytechnic University named after Peter the Great, Russia); Sergey V. Zavjalov (Peter the Great St. Petersburg Polytechnic University, Russia)

13:20 – 13:30 Q&A for Session VII: ZOOM session

13:30 – 15:10

Section VIII: Section II: Signals and PHY II

Electromagnetic waves propagation in low-profile SIW structures

Ekaterina Kiseleva, Artem Galushko and Alexander Sochava (Peter the Great St. Petersburg Polytechnic University, Russia)

Observation interval analysis for Faster-than-Nyquist signals coherent detection with decision feedback

Ilya Lavrenyuk and Sergey B. Makarov (Peter the Great St. Petersburg Polytechnic University, Russia); Wei Xue (Harbin Engineering University, China)

Section IX: Networks and correspondent issues

Heuristic Design Algorithm for Scheduling of URLLC and eMBB Traffics in 5G Cellular Networks

Jerzy Martyna (Jagiellonian University, Poland)

Geometrical approach to the plane tessellation in the IEEE 802.11 networks channel planning

Anton Vikulov and Alexander Paramonov (The Bonch-Bruевич Saint-Petersburg State University of Communication, Russia); Tatiana M Tatarnikova (Russian State Hydrometeorological University & State University of Aerospace Instrumentation, Russia)

Advancement of fingerprint polarimetric scheme for purposes of authentication

Pavel Trubin, Aleksandr A Murashov and Dmitry Suntsov (Peter the Great St. Petersburg Polytechnic University (SPbPU), Russia); Elena Velichko (Peter the Great St. Petersburg Polytechnic University, Russia)

Network slice degradation probability as a metric for defining slice performance isolation

Nikita Polyakov (RUDN University, Russia); Natalia Yarkina (Peoples' Friendship University of Russia, Russia); Konstantin Samouylov (Peoples' Friendship University of Russia, Russia)

Using a Machine Learning Model for Malicious URL Type Detection

Suet Ping Tung and Ka Yan Wong (HKU School of Professional and Continuing Education, Hong Kong); Ievgeniia Kuzminykh (King's College London, United Kingdom (Great Britain) & Kharkiv National University of Radio Electronics, Ukraine); Taimur Bakhshi (National University of Computer and Emerging Sciences, Pakistan); Bogdan Ghita (University of Plymouth & Centre for Security, Communications, and Network Research, United Kingdom (Great Britain))

16:20 – 16:30 Q&A for Sessions VIII and IX: ZOOM session and conference closing remarks